Uniform estimate for the smooth part of the multiplicative convolution sum

Tony Haddad, Dimitris Koukoulopoulos

doi:10.5802/jep.317; CC BY 4.0

Editorial note. The counted statement is the author's unlabelled lem environment at source0.tex lines 1394-1399; its proof environment opens at line 1401, the line after the blank separator, so the pairing is by adjacency and the retained proof range 1401-1446 is the whole environment.

Editorial note. Three environments are retained as context because the counted proof depends on them: the definition of the class of multiplicative functions with its labels (source lines 271-290), the description of the function on square-free arguments (source lines 294-297) and the partial summation bounds prel:partial summation bounds (source lines 1678-1685).

Editorial note. Because the lemma carries no author label, its identity is recorded by the pinned source position of the two environments (statement lines 1394-1399, proof lines 1401-1446) and by the claim key, which encodes that position.

Editorial note. This article carries no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the original printed number of this unlabelled lemma is not recoverable from the pinned material and is therefore not asserted.

Editorial scope. One original result of the pinned publisher article, counted once: the uniform estimate for the smooth part of the convolution sum (unlabelled lemma at source0.tex lines 1394-1399).

Allowed standard prerequisites (editorial).

- The class of multiplicative functions with a size condition. The functions of the class are multiplicative, and their values on square-free arguments are determined by the exponents attached to the prime factors, with a size condition on those exponents.. It is the class to which the dominated function belongs.
- Square-full and square-free decomposition. Every integer factors uniquely as a product of its square-full part and a square-free part, and this decomposition is used to split the divisors.. It is the first step of the proof.
- Partial summation. Sums over integers of a multiplicative function can be evaluated by partial summation once the corresponding summatory function is known.. It converts the estimate obtained for the sum into the stated bound.

Definition 0.1 (The class of functions $\mathcal{F}_k(\boldsymbol{\alpha})$)**.** Given $k \in \mathbb{Z}_{\geq 2}$ and $\boldsymbol{\alpha} \in \Delta^{k-1}$, let $\mathcal{F}_k(\boldsymbol{\alpha})$ be the set of functions $f\colon \mathbb{N}^k \to \mathbb{R}_{\geq 0}$ satisfying the following three properties:

a. We have $\sum_{d_1 \cdots d_k = n} f(d_1, \dots, d_k) = 1$ for all $n \in \mathbb{N}$. Thus, for each fixed $n \in \mathbb{N}$, the function $f(\boldsymbol{d})$ defines a probability mass function on the set of vectors $\boldsymbol{d} \in \mathbb{N}^k$ satisfying $d_1 \cdots d_k = n$.

b. Whenever $\boldsymbol{d}$ satisfies
$$d_i \coloneqq \begin{cases} p & \text{if } i = j, \\ 1 & \text{if } i \neq j. \end{cases}$$
for some $1 \leq j \leq k$ and prime p, then $f(\boldsymbol{d}) = \alpha_j$.

c. The function f is *multiplicative*, i.e. for any vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{N}^k$ with $a_1 \cdots a_k$ and $b_1 \cdots b_k$ being co-prime, we have the property
$$f(a_1 b_1, \dots, a_k b_k) = f(a_1, \dots, a_k) \cdot f(b_1, \dots, b_k).$$

$$f(d_1, \dots, d_k) = \prod_{j=1}^{k} \alpha_j^{\omega(d_j)}. \tag{1}$$

Proposition 0.2. *Uniformly for $\alpha \in [0,1]$ and $x, y \geq 2$, we have*
$$\sum_{n \leq x} \alpha^{\omega(n)} \ll x(\log x)^{\alpha - 1} \quad \text{and} \quad \sum_{x/y < n \leq xy} \frac{\alpha^{\omega(n)}}{n} \ll (\log y)(\log(xy))^{\alpha - 1}.$$

Lemma 0.3. *There exists a universal constant $x_0 \geq 2$ such that if $x \geq x_0$ and $z \in [2, (\log x)^4]$, then we have the uniform estimate*
$$S(z) \ll \frac{\log z}{\sqrt{z}} \cdot \frac{x}{(1 + u \log x)^{1-\alpha}(1 + (1-u)\log x)^{\alpha}}.$$

Proof. If $d_1 \cdots d_k = n$, then we may uniquely write $d_j = d_j' d_j''$ with $d_j' | n^\flat$ and $d_j'' | s(n)$. In particular, $d_j'' \leq s(n) \leq x/n^\flat \leq z$ for all j. Hence, if $d_i \in [x^u z^{-3}, x^u z^3]$, we must also have that $d_i' \in [x^u z^{-4}, x^u z^3]$. Hence, using property (c) of Definition 0.1, we deduce that
$$\sum_{\substack{d_1 \cdots d_k = n \\ d_i \in [x^u z^{-3}, x^u z^3]}} f(d_1, \dots, d_k) \leq \sum_{\substack{d_1' \cdots d_k' = n^\flat \\ d_i' \in [x^u z^{-4}, x^u z^3]}} f(d_1', \dots, d_k') \sum_{d_1'' \cdots d_k'' = s(n)} f(d_1'', \dots, d_k'').$$

Relation (1) implies that $f(d_1', \dots, d_k') = \prod_{j=1}^k \alpha_j^{\omega(d_j')}$. Moreover, using property (a) of Definition 0.1, we find that the sum over the d_j'''s equals 1. In conclusion,
$$\begin{aligned} \sum_{\substack{d_1 \cdots d_k = n \\ d_i \in [x^u z^{-3}, x^u z^3]}} f(d_1, \dots, d_k) &\leq \sum_{\substack{d_1' \cdots d_k' = n^\flat \\ d_i' \in [x^u z^{-4}, x^u z^3]}} \prod_{j=1}^{k} \alpha_j^{\omega(d_j')} \\ &= \sum_{\substack{d | n^\flat \\ d \in [x^u z^{-4}, x^u z^3]}} \alpha^{\omega(d)} (1-\alpha)^{\omega(n^\flat/d)}. \end{aligned}$$

Let $n^\flat = dm$ and $s(n) = b$. If $n^\flat \in [x/z, 2x/z]$ and $d \in [x^u z^{-4}, x^u z^3]$, then $m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]$. Hence, we find that

$$\begin{aligned} S(z) &\leq \sum_{\substack{dm \in [x/z, 2x/z] \\ d \in [x^u z^{-4}, x^u z^3] \\ m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]}} \sum \alpha^{\omega(d)} (1-\alpha)^{\omega(m)} \sum_{\substack{b \leq z \\ b \text{ square-full}}} 1 \\ &\ll \sqrt{z} \sum_{\substack{dm \in [x/z, 2x/z] \\ d \in [x^u z^{-4}, x^u z^3] \\ m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]}} \sum \alpha^{\omega(d)} (1-\alpha)^{\omega(m)} = \sqrt{z} \cdot (S_1 + S_2), \end{aligned} \tag{2}$$

where S_1 denotes the double sum over d and m with the additional constraint $d \leq \sqrt{2x/z}$, and S_2 is the corresponding sum over pairs (d, m) with $d > \sqrt{2x/z}$ and $m \leq \sqrt{2x/z}$.

First, we bound S_1. Note that for the conditions on d to be compatible we must have that $u \leq 2/3$, provided that x is large enough. Assuming this is the case, we apply twice Proposition 0.2 to find that

$$\begin{aligned} S_1 &\leq \sum_{\substack{d \leq \sqrt{2x/z} \\ d \in [x^u z^{-4}, x^u z^3]}} \alpha^{\omega(d)} \sum_{m \leq 2x/dz} (1-\alpha)^{\omega(m)} \\ &\ll \sum_{\substack{d \leq \sqrt{2x/z} \\ d \in [x^u z^{-4}, x^u z^3]}} \alpha^{\omega(d)} \cdot \frac{x (\log x)^{-\alpha}}{dz} \\ &\ll \frac{x (\log z)(1 + u \log x)^{\alpha - 1} (\log x)^{-\alpha}}{z}. \end{aligned} \tag{3}$$

We bound S_2 in a similar manner. For this sum to be non-empty, we have $u \geq 1/3$. If this is the case, then we have

$$\begin{aligned} S_2 &\leq \sum_{\substack{m \leq \sqrt{2x/z} \\ m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]}} (1-\alpha)^{\omega(m)} \sum_{d \leq 2x/mz} \alpha^{\omega(d)} \\ &\ll \sum_{\substack{m \leq \sqrt{2x/z} \\ m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]}} (1-\alpha)^{\omega(m)} \cdot \frac{x (\log x)^{\alpha - 1}}{mz} \\ &\ll \frac{x (\log z)(1 + (1-u) \log x)^{-\alpha} (\log x)^{\alpha - 1}}{z}. \end{aligned} \tag{4}$$

Combining (2), (3) and (4), while keeping in mind that $S_1 = 0$ if $u > 2/3$ and that $S_2 = 0$ if $u < 1/3$, completes the proof of the lemma, □

References

[1] R. Arratia - "On the central role of scale invariant Poisson processes on (0,∞)", in Microsurveys in discrete probability (Princeton, NJ, 1997), DIMACS Ser. Discrete Math. Theoret. Comput. Sci., vol. 41, American Mathematical Society, Providence, RI, 1998, p. 21-41

[2] R. Arratia - "On the amount of dependence in the prime factorization of a uniform random integer", in Contemporary combinatorics, Bolyai Soc. Math. Stud., vol. 10, János Bolyai Math. Soc., Budapest, 2002, p. 29-91

[3] R. Arratia, A. D. Barbour & S. Tavaré - "A tale of three couplings: Poisson-Dirichlet and GEM approximations for random permutations", Combin. Probab. Comput. 15 (2006) no. 1-2, p. 31-62

[4] G. Bareikis & E. Manstavičius - "On the DDT theorem", Acta Arith. 126 (2007) no. 2, p. 155-168

[5] P. Billingsley - "On the distribution of large prime divisors", Period. Math. Hungar. 2 (1972), p. 283-289

[6] R. de la Bretèche & G. Tenenbaum - "Sur les processus arithmétiques liés aux diviseurs", Adv. in Appl. Probab. 48 (2016) no. A, p. 63-76

[7] J.-M. Deshouillers, F. Dress & G. Tenenbaum - "Lois de répartition des diviseurs. I", Acta Arith. 34 (1979) no. 4, p. 273-285 (loose errata)

[8] P. Donnelly & S. Tavaré - "The population genealogy of the infinitely-many neutral alleles model", J. Math. Biol. 25 (1987) no. 4, p. 381-391

[9] S. Feng - The Poisson-Dirichlet distribution and related topics. Models and asymptotic behaviors, Probability and its Appl., Springer, Heidelberg, 2010

[10] N. L. Johnson, S. Kotz & N. Balakrishnan - Discrete multivariate distributions, Wiley Series in Probab. and Stat.: Applied Probab. and Stat., John Wiley & Sons, Inc., New York, 1997

[11] J. F. C. Kingman - Poisson processes, Oxford Studies in Probab., vol. 3, The Clarendon Press, Oxford University Press, New York, 1993

[12] S. Kotz, N. Balakrishnan & N. L. Johnson - Continuous multivariate distributions. Vol. 1. Models and applications, Wiley Series in Probab. and Stat.: Applied Probab. and Stat., Wiley-Interscience, New York, 2000

[13] D. Koukoulopoulos - The distribution of prime numbers, Graduate Studies in Math., vol. 203, American Mathematical Society, Providence, RI, 2019

[14] G. Last & M. Penrose - Lectures on the Poisson process, Institute of Math. Stat. Textbooks, vol. 7, Cambridge University Press, Cambridge, 2018

[15] S.-K. Leung - "Dirichlet law for factorisation of integers, polynomials and permutations", Math. Proc. Cambridge Philos. Soc. 175 (2023) no. 3, p. 649-676

[16] S.-K. Leung - "Dirichlet law with arbitrary parameters for factorizations of integers" (2025), Retrieved on 24 August 2025 from https://sites.google.com/view/sunkaileung/miscellaneous

[17] H. L. Montgomery & R. C. Vaughan - Multiplicative number theory. I. Classical theory, Cambridge Studies in Advanced Math., vol. 97, Cambridge University Press, Cambridge, 2007

[18] S. Nyandwi & A. Smati - "Distribution laws of pairs of divisors", Integers 13 (2013), article ID A13, 13 pages

[19] G. Tenenbaum - "A rate estimate in Billingsley's theorem for the size distribution of large prime factors", Q. J. Math. 51 (2000) no. 3, p. 385-403